THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. If $a = 9$ and $b = 4$, the value of $3a - 5b + 8$ is: 1

(a) 15 **(b)** 16 **(c)** 30 **(d)** 14

2. A rhombus has side 15 cm and one diagonal 18 cm. Its area is: 1

(a) 108 cm² **(b)** 216 cm² **(c)** 270 cm² **(d)** 60 cm²

3. $\sqrt{1.44}$ is equal to: 1

(a) 0.12 **(b)** 1.2 **(c)** 1.3 **(d)** 12.0

4. If $(2^3)^x = 2^{12}$, then x is: 1

(a) 36 **(b)** 9 **(c)** 15 **(d)** 4

5. The largest 3-digit number in base 2, written in decimal, is: 1

(a) 3 **(b)** 5 **(c)** 7 **(d)** 8

6. In a rhombus ABCD, $\angle A = (4x + 10)^\circ$ and $\angle B = (5x + 17)^\circ$. The value of x is: 1

(a) 17 **(b)** 21 **(c)** 16 **(d)** 18

7. The smallest number divisible by 4, 6 and 10 is: 1

(a) 64 **(b)** 30 **(c)** 120 **(d)** 60

8. The value of $(x + 3)^2$ when $x = 7$ is: 1

(a) 58 **(b)** 79 **(c)** 100 **(d)** 106

9. 6 pipes fill a tank in 10 hours. 4 such pipes will fill it in: 1

(a) 15 hours **(b)** 60 hours **(c)** 12 hours **(d)** 18 hours

10. The simple interest on a sum at 5% p.a. for 4 years is ₹ 600. The sum (in ₹) is: 1

(a) 3000 **(b)** 3300 **(c)** 6000 **(d)** 2700

11. Two vertical poles of heights 40 m and 22 m stand 24 m apart on level ground. The distance between their tops (in m) is: 1

(a) 42 **(b)** 30 **(c)** 28 **(d)** 31

12. A sum of ₹ 1200 buys 24 kg of rice at ₹ 50 per kg. When the price falls to ₹ 40 per kg, the same money buys (kg): 1

(a) 30 **(b)** 48 **(c)** 28 **(d)** 24

13. A polyhedron has 12 faces and 30 edges. Its number of vertices is: 1

(a) 40 **(b)** 44 **(c)** 18 **(d)** 20

14. The mean of 5 numbers is 28. The number to be added so that the mean becomes 30 is: 1

(a) 45 **(b)** 2 **(c)** 40 **(d)** 30

15. A rectangle has length $(3x + 2)$ cm and breadth $(2x + 5)$ cm. Its perimeter (cm) is: 1

(a) $10x + 7$ **(b)** $6x^2 + 10$ **(c)** $5x + 7$ **(d)** $10x + 14$

16. A path 2 m wide runs around the outside of a 20 m by 15 m rectangular garden. The area of the path is: 1

(a) 140 m² **(b)** 456 m² **(c)** 156 m² **(d)** 16 m²

17. $\sqrt[3]{0.027}$ is equal to: 1

(a) 0.3 **(b)** 0.4 **(c)** 3.0 **(d)** 0.03

18. $2^3 \times 5^3$ equals: 1

(a) 1000 **(b)** 250 **(c)** 30 **(d)** 133

19. The number of digits in the binary form of 20 is: 1

(a) 7 **(b)** 4 **(c)** 6 **(d)** 5

20. In a kite ABCD with AB = AD and CB = CD, $\angle A = 60^\circ$ and $\angle C = 130^\circ$. The angle $\angle B$ is: 1

(a) 85° **(b)** 75° **(c)** 95° **(d)** 170°

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Expand and simplify $2(x + 5) + 3(x - 1)$. 2

22. A rectangle has perimeter 80 cm and length 26 cm. Find its breadth and area. 2

23. Evaluate $\sqrt[3]{\dfrac{27}{512}}$. 2

24. Find x if $2^x = \dfrac{1}{32}$. 2

25. Write CXXXIX + CCLXII as a Roman numeral. 2

26. The diagonals of a parallelogram ABCD meet at O with AO $= (3x + 5)$ cm and OC $= (2x + 9)$ cm. Find x and AC. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Multiply $101_2 \times 11_2$ and give the answer in binary. 3

28. The adjacent sides of a parallelogram are in the ratio 2:5 and its perimeter is 56 cm. Find the lengths of its sides. 3

29. Find the digit * so that 83*5 is divisible by 11. Show your working. 3

30. Find 197^2 using an algebraic identity. 3

31. A : B = 1 : 2 and B : C = 6 : 5. If A + B + C = 150, find the three numbers. 3

OR

In how many years will ₹ 7500 amount to ₹ 9300 at 8% simple interest per annum? 3

32. Show that the triangle with sides 18 cm, 24 cm and 30 cm is right-angled and find its area. 3

OR

A:B = 1:2 and B:C = 3:4. If A+B+C = 170, find the value of each. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Find the smallest number which leaves remainder 3 when divided by 6, 8 and 12. 5

34. Use identities to evaluate: 5

(i) 103^2

(ii) 88×92

(iii) $76^2 - 24^2$.

35. Anil and Bhanu start a business with ₹ 15000 and ₹ 25000. They share a profit of ₹ 800 in the ratio of their investments. Find the ratio of investments and each person's share. 5

36. A man deposits ₹ 4000 at simple interest and another ₹ 6000 at the same rate. After 2 years the total interest is ₹ 1000. Find the rate and the interest earned on each sum. 5

OR

An isosceles triangle has equal sides of 40 cm and base 48 cm. Find its height, area and perimeter. 5

37. 12 workers can complete a task in 20 days. After 8 days, 6 more workers join them. In how many more days will the work be finished? 5

OR

A Koch snowflake is built from an equilateral triangle of side 9 cm. Find its perimeter at stage 0, stage 1 and stage 2. 5

38. A class of 20 students has a mean mark of 50. Another 30 students with mean 60 join it. Find the mean of the combined group and the change in the original mean. 5

OR

Think of a two-digit number with digits a and b. Add its reverse. Show that the sum is 11(a+b) and find the sum for the digits 3, 9. Is the sum divisible by 11? 5

END OF QUESTION PAPER Total: 80 Marks

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